TEXORA · SESSIONAL NOTES · STUDY SHEET

Practices on CAD/CAM Systems for Apparel Manufacturing

Working the digital chain in the lab: pattern creation and grading in CAD, marker planning with a measured efficiency, and the CAM outputs — plotter, cutter and data for the sewing floor.

AM-4202 · ADVANCED APPAREL MANUFACTURING-II SESSIONAL · STUDY 4 · LEVEL 4 TERM 2

1. Aim and place of the study

Aim: to operate the apparel CAD system through its working cycle — digitize or draft a pattern, grade it to a size range, plan a marker and read its efficiency — and to state how the CAM side (plotting, automated cutting, production data) consumes the CAD output.

2. Equipment and materials

Equipment: CAD workstation with apparel pattern software (pattern design, grading and marker modules), digitizing tablet or scanner for paper patterns, plotter, and the lab's link to the cutting room. **Specimen:** the men's basic trouser of TE-3108 (size M: waist 82, hip 102, outseam 104, inseam 78; grades 5 cm at waist and hip, 2 cm at thigh, knee and hem) — its grading rules are already established, so the study can concentrate on the system rather than the garment.

3. The CAD working cycle

1. **Pattern entry.** Either draft the pattern directly on screen from the measurement chart, or digitize the approved paper pattern: place it on the tablet, register the points in sequence, and let the software rebuild each piece as a closed curve with its internal marks (grain line, notches, drill holes).
2. **Verification.** Walk the seams on screen: the software checks that joining seams match in length (front and back side seams, inseam pairs) and reports any mismatch before a single piece is cut — the digital equivalent of the paper pattern check.
3. **Grading.** Apply the grade rules at each grade point: for the trouser, waist and hip points move 5 cm between sizes, thigh, knee and hem 2 cm, outseam by its length grade. The system generates S, M,

L, XL from the M base; the student inspects the graded nest for distorted shapes — grading multiplies any base-pattern error into every size.

4. **Marker planning.** Load the size ratio for the order, set the fabric width, and place the pieces: grain-parallel, matched for the fabric's behaviour, large pieces first. The software reports marker length and efficiency continuously as pieces are placed; automatic nesting can be run and then improved by hand.
5. **Output.** Send the marker to the plotter for a paper lay or directly to the automatic cutter; export the consumption and piece data to the production file used by cutting and costing (Studies 5 and 6).

4. The marker made in this study

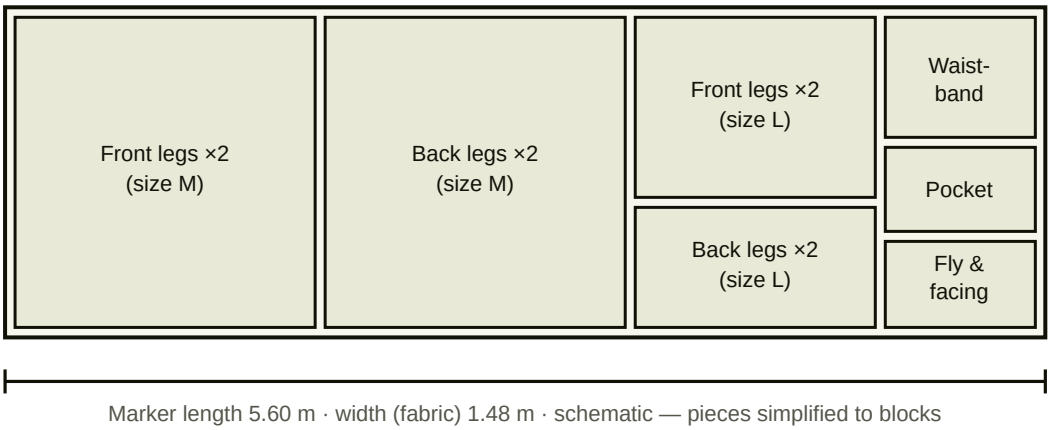


Figure 1 — Marker plan as the CAD system reports it: every piece inside the fabric width, grain lines parallel, no overlaps. Efficiency is computed from this placement, not estimated.

5. Worked example: marker efficiency and consumption

Marker efficiency compares the area of the pattern pieces with the area of fabric the marker consumes:

$$\text{efficiency} = (\text{total pattern-piece area} \div \text{marker area}) \times 100$$

For the study marker the software reports total piece area **6.72 m²** for a marker **5.60 m** long at the set width of **1.48 m**:

$$\text{marker area} = 5.60 \times 1.48 = 8.288\ \text{m}^2 \rightarrow \text{efficiency} = 6.72 \div 8.288 \times 100 = 81.1\%$$

An efficiency of 81.1% means 18.9% of the spread fabric is cutting waste (selvedge loss, piece gaps, end loss) — the single largest fabric-cost lever in the factory, which is why marker work is a skilled job. The same marker data gives consumption directly: the marker holds 4 trousers (2 M + 2 L), so fabric per garment = 5.60 ÷ 4 = **1.40 m** at this width and size mix — note carefully that this differs

from the 1.12 m figure of Study 3, which was for a narrower size ratio at 150 cm width; consumption is a property of *this* marker, never of the garment alone.

> **WHAT CAD CHANGES — AND WHAT IT DOES NOT**
>
> CAD removes drafting arithmetic, grading labour and marker guesswork, and it makes every result repeatable and auditable. It does not remove judgement: a wrong base pattern grades beautifully into four wrong sizes, and an efficient marker on the wrong width or ratio is still the wrong marker. The checks of §3 — seam walking, graded-nest inspection, efficiency review — are the human content of the study.

6. The CAM side, in brief

Plotting turns the marker into a full-size paper lay for manual cutting rooms and into mini-markers for approval. **Automated spreading and cutting** consume the same file: the spreader lays the commanded number of plies, and the CNC knife (or laser for some synthetics) cuts the pieces with notches and drill marks, at speeds and accuracies manual cutting cannot match — with vacuum hold-down keeping the lay still. **Data flow** is the quiet third output: piece counts, consumption and marker records pass to production planning and costing, so the cutting room, the line (Study 5) and the cost sheet (Study 5) all work from one set of numbers.

Precautions

- Save versioned files at every stage (base, graded, marker); an overwritten base pattern cannot be recovered and invalidates every dependent file.
- Confirm fabric width and size ratio on the order sheet before nesting — the two settings that silently change every output number.
- Inspect the graded nest on screen before plotting: grading errors are cheap here and expensive after cutting.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. A marker 6.20 m long at 1.50 m width carries pieces totalling 7.44 m². Find the efficiency.
2. Why can two correct markers for the same garment give different consumptions per garment?
3. What does "walking the seams" check, and why is it done before grading?
4. State the three CAM outputs of the CAD file.

Answer key

1. Marker area = 6.20 × 1.50 = 9.30 m². Efficiency = 7.44 ÷ 9.30 × 100 = 80.0%.
2. Consumption depends on the marker: fabric width, the size ratio placed, and how efficiently the pieces nest. Change any of these and the metres per garment change, although the garment and its patterns are identical.
3. It checks that seams which will be sewn together are equal in length (allowing only the designed ease), so the pieces will join without forcing. It is done on the base pattern because grading copies the base into every size — a mismatch found after grading must be corrected in all sizes.
4. The plotted marker (paper lay and mini-markers), the cutter file driving automated spreading/cutting, and the production data set (pieces, consumption, marker records) used by planning and costing.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib